SUCCESSOR TRANSFER ON CANONICALLY ORDERED SET PARTITIONS

ANONYMOUS

Abstract. Order the blocks of a set partition by their minima. In one successor- transfer step, every nonsingleton block simultaneously sends its maximum to the next block around a directed cycle. We determine the finite dynamics of this literal rule. On partitions of $[n]$ into k blocks, the maximum tail is $\min\{n-2, 2k-2\}$ for $1 < k < n$; the one-block and all-singleton states are fixed, and the global maximum is $n-2$. Recurrent states have explicit dense and sparse prefix–suffix forms, giving their exact counts, and every nontrivial recurrent k-block state has period k. Independently, the one-step fibre of every target is the trace of a product of explicit 5×5 matrices. Its states record whether an incoming label is absent, the sole element, the minimum, the maximum, or an interior element of a target block. This trace is sensitive to labelled interlacing: the targets $025 \mid 134$ and $035 \mid 124$ have identical block size/minimum/maximum data but fibres two and one. Classical carrier and load mechanisms are treated as background. External circulation remains on hold.

1. The literal rule and the theorem package

Let $[n] = \{0, 1, \ldots, n-1\}$, and let $\Pi_{n,k}$ be the set of its set partitions into k nonempty blocks. We always write

$$\pi = (B_0, B_1, \ldots, B_{k-1}), \qquad \min B_0 < \min B_1 < \cdots < \min B_{k-1}.$$

Define $\mathcal{T} : \Pi_{n,k} \to \Pi_{n,k}$ as follows. Every block with at least two elements removes its maximum and sends that element to B_{i+1}, with subscripts modulo k; singleton blocks send nothing. All removals and additions occur simultaneously. For $k = 1$, removal and return occur in the same block and $\mathcal{T}$ is the identity.

The canonical ordering in this definition is not an update-dependent relabeling. Indeed, each block retains its minimum. Every element of the new block $i+1$ is either from B_{i+1} or is $\max B_i$, hence is strictly larger than $\min B_i$, whereas the new block i has minimum at most $\min B_i$. Thus adjacent output minima remain strictly increasing. This argument includes the donation $B_{k-1} \to B_0$: it affects the first block but creates no last-to-first order comparison.

The restricted-growth word of π is $w = w_0 \cdots w_{n-1}$, where $w_j = i$ exactly when $j \in B_i$. Its first occurrences appear in the order $0, 1, \ldots, k-1$ [8]. The literal rule becomes

$$\boxed{\text{for every repeated letter } i\text{, replace its final occurrence by } i+1 \pmod{k}\text{, simultaneously.}} \tag{1}$$

Every donating letter retains its original first occurrence. If replacing the last i by $i+1$ creates an earlier first occurrence of $i+1$, that position is still after the first i; the wrap to zero is harmless. Hence (1) remains a restricted-growth word on the same k letters.

Restricted-growth encodings and their Stirling enumeration are background [8]. Whirling already supplies sequential local actions on restricted-growth functions [4]; it is not (1). The excess-load projection below is exactly a threshold-one parallel chip-firing rule on a directed cycle and is likewise background [3]. We also distinguish the literal rule from Bulgarian solitaire [1], jeu-de-taquin promotion and rowmotion [5, 6], box-ball dynamics [7], and set-partition stack sorting [2]. None of these carrier, load, sorting, counting, or generic matrix mechanisms is counted as part of the residual theorem package.

For a point in a finite self-map, let $\operatorname{tail}(\pi)$ be its least entrance time into a cycle. Put $m = n - k$, let $\left\{ {m \atop k} \right\}$ denote a Stirling number of the second kind, and write $(k)_m = k!/(k-m)!$.

Theorem 1 (sharp clocks and recurrent strata)**.** *The successor-transfer map has the following complete temporal behavior.*

(i) If $1 < k < n$, then

$$\max_{\pi \in \Pi_{n,k}} \operatorname{tail}(\pi) = \min\{n-2,\, 2k-2\}. \tag{2}$$

For $k = 1$ and $k = n$, the unique state is fixed. Consequently, for every $n \geq 2$, the maximum over all set partitions of $[n]$ is $n - 2$.

(ii) In the dense range $n \geq 2k$, a k-block state is recurrent precisely when its first $n - k$ letters form a surjective restricted-growth word on all k letters and its final k letters are a permutation of them. In the sparse range $n \leq 2k$, it is recurrent precisely when its first k letters are $01 \cdots (k-1)$ and its final $n - k$ letters are distinct.

(iii) The number $R_{n,k}$ of recurrent k-block states is

$$R_{n,k} = \begin{cases} k!\left\{ {n-k \atop k} \right\}, & n \geq 2k, \\ (k)_{n-k}, & n \leq 2k. \end{cases} \tag{3}$$

At $n = 2k$, both descriptions read $01 \cdots (k-1)$ followed by a permutation, and both counts equal $k!$. Every recurrent state with $1 < k < n$ has exact period k. The cases $k = 1$ and $k = n$ have period one. Thus for $n \geq 2$ the possible periods over all strata are exactly $1, 2, \ldots, n-1$; at $n = 1$ the unique state is fixed.

2. The directed-cycle load factor

Let c_i be the multiplicity of letter i and set

$$z_i = c_i - 1 \geq 0, \qquad \sum_{i=0}^{k-1} z_i = m.$$

Writing $b_i = \mathbf{1}_{\{z_i>0\}}$, (1) gives

$$z_i(t+1) = z_i(t) - b_i(t) + b_{i-1}(t), \qquad i \pmod k. \tag{4}$$

This entire factor is the directed-cycle chip-firing mechanism already subtracted above. We record its finite-time lemma because the labelled lift needs an exact entrance bound.

Lemma 2 (queue smoothing)**.** *Assume $m > 0$. If $m \leq k$, then every z_i is in $\{0, 1\}$ after at most $m-1$ steps. If $m \geq k$, then every z_i is positive after at most $k-1$ steps. The resulting sparse and dense regimes are forward invariant.*

Proof. Lift indices periodically to $\mathbb{Z}$. Choose integers $H_i(0)$ with

$$H_i(0) - H_{i-1}(0) = z_i(0), \qquad H_{i+k}(0) = H_i(0) + m.$$

Since z_i is a nonnegative integer, (4) is equivalent to

$$H_i(t+1) = \max\{H_i(t) - 1, H_{i-1}(t)\}. \tag{5}$$

Induction on t yields the explicit solution

$$H_i(t) = \max_{0 \leq r \leq t} \big(H_{i-r}(0) - (t-r)\big). \tag{6}$$

Fix i, t and put $X_r = H_{i-r}(0) + r$. Taking the difference at adjacent indices in (6) gives

$$z_i(t) = 1 + \max_{0 \leq r \leq t} X_r - \max_{1 \leq r \leq t+1} X_r. \tag{7}$$

The two maxima share $X_1, \ldots, X_t$. If $z_i(t) = 0$, the second maximum must be the endpoint X_{t+1} and exceed the first by one; hence

$$\sum_{r=0}^{t} z_{i-r}(0) = H_i(0) - H_{i-t-1}(0) \leq t.$$

If $z_i(t) \geq 2$, the first maximum must instead be X_0 and exceed the second by at least one; hence

$$\sum_{r=0}^{t} z_{i-r}(0) \geq t + 2. \tag{8}$$

For $m \leq k$, take $t = m - 1$. A surviving value $z_i(t) \geq 2$ would force at least $m+1$ units in a cone although the total mass is m. For $m \geq k$, take $t = k-1$. An empty $z_i(t)$ would bound the mass in one complete cycle by $k-1$, contradicting $m \geq k$. Finally, a binary load obeys $z_i(t+1) = z_{i-1}(t)$, while an everywhere-positive load is fixed under (4); both regimes are invariant. □

3. Labelled windows, recurrence, and sharpness

Load smoothing does not yet make the labelled partition recurrent. A second phase orders a distinguished word window.

Lemma 3 (canonical windows)**.** *After Lemma 2, at most $k-1$ further steps put the state in one of the following invariant forms.*

(a) If $m \geq k$, the final k word positions form a permutation of all k letters.
(b) If $m \leq k$, the first k word positions are $01 \cdots (k-1)$.

Proof. In the dense regime every letter remains repeated. Reverse the word and inspect its first k positions, equivalently the original final k positions. For every colour present in this window, exactly its first reversed occurrence moves one site clockwise under (1); an absent colour contributes no moving window token. The occupancy vector therefore follows (4), now with mass exactly k. The dense case of Lemma 2 makes all k occupancies positive by time $k-1$, hence each is one.

In the sparse regime every full-word multiplicity is one or two. Let q_i count letter i in the first k positions. Then $q_i \in \{0,1,2\}$ and $\sum_i q_i = k$. A value two has both copies in the prefix and moves its later copy clockwise; a value at most one does not move a prefix token. Thus

$$q_i' = q_i - \mathbf{1}_{\{q_i=2\}} + \mathbf{1}_{\{q_{i-1}=2\}}. \tag{9}$$

Regard each 2 as one excess particle and each 0 as a hole. Particles move one site per step through occupied sites, preserve cyclic order, and annihilate on the first unmatched hole. Pairing particles with holes in cyclic order on the universal cover shows that every pair meets within $k-1$ steps. Then all $q_i = 1$. A restricted-growth word whose first k letters are a permutation of $0,\dots,k-1$ must have those letters in the order $01\cdots(k-1)$.

In the dense form, every last occurrence lies in the final permutation, so the prefix is fixed and the permutation is incremented modulo k. In the sparse form, the final m letters are exactly the duplicated letters, are distinct, and are incremented modulo k while the canonical prefix is fixed. Both forms are invariant. □

Proof of Theorem 1(ii)–(iii). A periodic state must already lie in the invariant smoothed-load regime; otherwise a forward iterate enters that regime and can never return. The same argument with Lemma 3 forces the appropriate canonical window. In the dense case, deleting the final permutation leaves a length-m prefix that is a restricted-growth word and contains every letter, because every full-word multiplicity is at least two. In the sparse case, the canonical prefix contains one copy of every letter and the remaining m letters are distinct. This proves necessity.

Conversely, each stated form is invariant by the last paragraph of the lemma. When $1 < k < n$, its nonempty final permutation or injection is increased componentwise modulo k, so the exact period is k. There are $\left\{ {m \atop k} \right\}$ surjective restricted-growth prefixes and $k!$ final permutations in the dense case. In the sparse case, the final m letters form an injection chosen in $(k)_m$ ways. This proves (3), including equality of the two formulas at $m = k$. The boundary strata were already observed to be fixed, and the period-set statement follows by ranging over $k = 2,\dots,n-1$. □

Proof of Theorem 1(i). Lemma 2 costs at most $\min\{m,k\}-1$ steps and Lemma 3 at most $k-1$. Their sum is $n-2$ when $m \le k$ and $2k-2$ when $m \ge k$, proving the upper bound in (2).

For sharpness, use the single restricted-growth word

$$w_{n,k} = 0^{m+1}12\cdots(k-1). \tag{10}$$

For $0 \le t \le \min\{m-1,k-1\}$, direct use of (1) gives

$$\mathcal{T}^t(w_{n,k}) = 0^{m+1-t}1^2 2^2 \cdots t^2 (t+1)(t+2)\cdots(k-1). \tag{11}$$

Thus load smoothing takes exactly $m-1$ steps if $m \le k$ and exactly $k-1$ steps if $m \ge k$.

Suppose first that $m \le k$. At the end of this load phase, the occupancy of the first k positions is

$$2^r 1^{k-2r} 0^r, \qquad r = \min\{m, \lfloor k/2 \rfloor\}.$$

Iteration of (9) shows by induction that its last coordinate remains zero for the first $k-1$ times and that the vector becomes 1^k at time $k-1$. Hence the labelled phase costs exactly $k-1$.

If $m \ge k$, write $d = m-k$. The load phase ends at $0^{d+2}1^2\cdots(k-1)^2$. For $1 \le s \le k-1$, another direct induction from (1) gives

$$\mathcal{T}^{k-1+s}(w_{n,k}) = 0^{d+1}12\cdots(s-1)s^2(s+1)^2\cdots(k-1)^2 01\cdots(s-1). \tag{12}$$

The terminal k positions first become a permutation at $s = k-1$. Therefore (10) has tail $n-2$ in the sparse range and $2k-2$ in the dense range.

For $n \ge 3$, choosing $k = n-1$ gives the global value $n-2$. At $n = 2$ both available strata are fixed and the same formula is zero. This completes the proof. □

4. An explicit five-state target-fibre trace

The load factor discards the label comparisons needed to invert one step. Fix a target $C = (C_0,\dots,C_{k-1})$ in canonical minimum order, initially with $k \ge 2$. If source block B_i donated, write $x_i = \max B_i$; then $x_i \in C_{i+1}$. If it was a singleton, write $x_i = \perp$. Any predecessor is forced to satisfy

$$B_i = \big(C_i \setminus \{x_{i-1}\}\big) \cup \{x_i\}, \tag{13}$$

where deletion or addition is omitted when its token is $\perp$.

We now print all local data needed to evaluate the matrices. Let $D = \{d_1 < \cdots < d_h\}$ be a nonempty target block and order the states as

$$\mathcal{S} = (\perp, \mathsf{s}, \mathsf{min}, \mathsf{max}, \mathsf{int}).$$

The candidate selected elements $E_a(D)$ are

a	$E_a(D)$
$\perp$	$\{\perp\}$
s	$\{d_1\}$ if $h = 1$, and $\varnothing$ otherwise
min	$\{d_1\}$ if $h \geq 2$, and $\varnothing$ otherwise
max	$\{d_h\}$ if $h \geq 2$, and $\varnothing$ otherwise
int	$\{d_2, \ldots, d_{h-1}\}$ if $h \geq 3$, and $\varnothing$ otherwise

After deleting a token of type a, let $r_a(D)$, $\ell_a(D)$, and $u_a(D)$ be the size, minimum, and maximum of the remainder. These data depend only on the state, even when the chosen interior element varies:

a	condition	$r_a(D)$	$\ell_a(D)$	$u_a(D)$
$\perp$	always	h	d_1	d_h
s	$h = 1$	0	–	–
min	$h \geq 2$	$h-1$	d_2	d_h
max	$h \geq 2$	$h-1$	d_1	d_{h-1}
int	$h \geq 3$	$h-1$	d_1	d_h

Rows failing their displayed condition are inadmissible. For a present outgoing state $b \neq \perp$, define the fully explicit threshold count

$$N_b(D; u) = \begin{cases} \mathbf{1}_{\{h=1,\ d_1 > u\}}, & b = \mathsf{s}, \\ \mathbf{1}_{\{h\geq 2,\ d_1 > u\}}, & b = \mathsf{min}, \\ \mathbf{1}_{\{h\geq 2,\ d_h > u\}}, & b = \mathsf{max}, \\ |\{j : 2 \leq j \leq h-1,\ d_j > u\}|, & b = \mathsf{int}. \end{cases} \tag{14}$$

For $0 \leq i < k$, rows refer to the type of x_{i-1} in C_i and columns to the type of x_i in C_{i+1}, with subscripts modulo k. For $i < k-1$, put $K_i(a, b) = 0$ unless both encoded remainders are admissible and nonempty; when they are, define

$$K_i(a, b) = \mathbf{1}_{\{\ell_a(C_i) < \ell_b(C_{i+1})\}}. \tag{15}$$

For the wrap index, put $K_{k-1}(a, b) = 1$; canonical block order has no last-to-first comparison. The explicit 5×5 matrix $M_i(C)$ is

$$\boxed{\begin{cases} (M_i)_{a,b} = 0, & a \text{ inadmissible or } r_a(C_i) = 0, \\ (M_i)_{a,\perp} = K_i(a, \perp)\mathbf{1}_{\{r_a(C_i)=1\}}, & r_a(C_i) > 0, \\ (M_i)_{a,b} = K_i(a, b) N_b(C_{i+1}; u_a(C_i)), & r_a(C_i) > 0,\ b \neq \perp. \end{cases}} \tag{16}$$

Equations (14)–(16) contain only block sizes, ordered labels, comparisons, and indicators, so no unspecified transition procedure remains.

Theorem 4 (every-target fibre)**.** *For every canonical k-block target C with $k \geq 2$,*

$$\boxed{|\mathcal{T}^{-1}(C)| = \operatorname{tr}\big(M_0(C) M_1(C) \cdots M_{k-1}(C)\big).} \tag{17}$$

Consequently, C lies in the image exactly when this trace is positive. For $k = 1$, the unique target has fibre one.

Proof. Fix a token choice. Removing incoming x_{i-1} from C_i gives the retained part of source block i. If $x_i = \perp$, then the source did not donate and must be a singleton; this is exactly the absent-token line of (16). If x_i is present, the retained part must be nonempty and x_i must strictly exceed its maximum, exactly the present-token line and the count (14). In either case the source minimum is the retained minimum. Its canonical order is therefore precisely the comparison in (15) for $i < k-1$. No comparison belongs after $i = k-1$, even though the donation itself wraps to block zero.

Conversely, a cyclic sequence of admissible states contributing to the trace, together with one of the locally counted label choices at each present step, reconstructs nonempty blocks by (13). The entry tests make every selected x_i exactly the maximum of a nonsingleton source block, make every absent selection an inactive singleton, and put the source blocks in canonical order. Applying $\mathcal{T}$ returns C. The selected maxima are recoverable from a predecessor, so this construction is bijective. Matrix multiplication joins each outgoing state to the next incoming state, and the trace closes $x_{k-1} \in C_0$. This proves (17). Positivity is the image criterion. $\square$

Singleton and wrap boundaries are visible in the matrices. Selecting the sole element of a singleton target block leaves $r_{\mathsf{s}} = 0$, so its entire outgoing row is zero. An all-singleton target instead has the unique all-$\perp$ state path of weight one. The matrix M_{k-1} omits the false last-to-first minimum comparison, while the trace still identifies its outgoing token type with the incoming type at C_0.

Here is a numerical evaluation that also isolates labelled interlacing. Use state order $(\perp, \mathsf{s}, \mathsf{min}, \mathsf{max}, \mathsf{int})$. For $A = 025 \mid 134$,

$$M_0(A) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_1(A) = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

whereas for $D = 035 \mid 124$,

$$M_0(D) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_1(D) = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Thus $\operatorname{tr}(M_0(A)M_1(A)) = 2$ and $\operatorname{tr}(M_0(D)M_1(D)) = 1$. The literal predecessors are

$$\mathcal{T}^{-1}(A) = \{023 \mid 145,\ 024 \mid 135\}, \qquad \mathcal{T}^{-1}(D) = \{034 \mid 125\}.$$

For both targets the ordered triples (block size, minimum, maximum) are $(3, 0, 5)$ and $(3, 1, 4)$. Hence the fibre is not determined by those coarse data; it retains interior label interlacing discarded by the temporal load factor.

5. Exact controls and scope

The paper-local standard-library verifier reconstructs the literal map and the matrices without importing any scouting implementation. It checks the restricted-growth update, stratum clocks, recurrent forms, exact periods, and counts for every set partition through $n = 10$; the max-plus cone identity and both implications on 532,467 bounded queue cases; the fibre trace against literal predecessor enumeration for all 26,442 targets through $n = 9$; the two printed matrix pairs; and the sharp family (10) in every nontrivial stratum through $n = 50$. Its frozen run records 1,217,025 exact assertions. These computations are falsification controls; the proofs above carry the all-parameter statements.

The note concerns only the simultaneous maximum-to-successor map. The five-state trace is a target-resolved structural formula, not a claim about a generic transfer-matrix method. The source search was bounded and confers no permission to circulate or submit the manuscript.

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References

[1] Jørgen Brandt. Cycles of partitions. *Proceedings of the American Mathematical Society*, 85(3):483–486, 1982. doi: 10.1090/S0002-9939-1982-0656129-5.

[2] Yunseo Choi, Katelyn Gan, Andrew Li, and Tiffany Zhu. On the set partitions that require maximum sorts through the *aba*-avoiding stack. *arXiv preprint arXiv:2403.05113*, 2024. doi: 10.48550/arXiv.2403.05113.

[3] David Ji, Michael Li, and Daniel Wang. Periods and atomic firing sequences of parallel chip-firing games on directed graphs. *Annals of Combinatorics*, 29(4):1155–1175, 2025. doi: 10.1007/s00026-025-00760-3.

[4] Michael Joseph, James Propp, and Tom Roby. Whirling injections, surjections, and other functions between finite sets. *Discrete Mathematics & Theoretical Computer Science*, 27(3), 2025. doi: 10.46298/dmtcs.14126.

[5] Marcel-Paul Schützenberger. Promotion des morphismes d'ensembles ordonnés. *Discrete Mathematics*, 2(1): 73–94, 1972. doi: 10.1016/0012-365X(72)90062-3.

[6] Jessica Striker and Nathan Williams. Promotion and rowmotion. *European Journal of Combinatorics*, 33(8): 1919–1942, 2012. doi: 10.1016/j.ejc.2012.05.003.

[7] Daisuke Takahashi and Junkichi Satsuma. A soliton cellular automaton. *Journal of the Physical Society of Japan*, 59(10):3514–3519, 1990. doi: 10.1143/JPSJ.59.3514.

[8] Michelle L. Wachs. σ-restricted growth functions and p, q-stirling numbers. *Journal of Combinatorial Theory, Series A*, 68(2):470–480, 1994. doi: 10.1016/0097-3165(94)90117-1.